Lab Exam 2 : Newton iterations for solving a nonlinear system

MTH 308: Numerical Analysis and Scientific Computing--I

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clc; clear;
% The nonlinear system F(x) = 0 is given by:
%   x1^3 + x1^2*x2 - x1*x3 + 6 = 0
%   exp(x1) + exp(x2) - x3 = 0
%   x2^2 - 2*x1*x3 - 4 = 0

% Solve F(x) = 0 using Newton's method
% Iterate until ||x^(k) - x^(k-1)||_inf < 1e-8

% Newton iteration:
% J(x^(k)) dx = F(x^(k)),   x^(k+1) = x^(k) - dx

x0 = [-1; -2; 1];
tol = 1e-8;
maxit = 50;

x = Newton_system(@f_fun, @Jacobian, x0, tol, maxit);
fprintf('\nFinal solution:\n');
fprintf('x = [% .6f, % .6f, % .6f]^T\n', x(1), x(2), x(3));

%-----------------------------Functions-----------------------------------%
function x = Newton_system(Ffun, Jfun, x0, tol, maxit)
    x = x0;
    err = 1.0;
    it = 0;

    fprintf('k\t   x1\t\t   x2\t\t  x3\t\t ||dx||_inf\t ||F(x)||_inf\n');

fprintf('-----------------------------------------------------------------------------------\n');

    while err > tol && it < maxit
        Fx = Ffun(x);
        Jx = Jfun(x);

        dx = Jx \ Fx;
        x = x - dx;

        err = max(abs(dx));
        it = it + 1;

        Fx = Ffun(x); % recompute after update

        fprintf('%d\t% .6f\t% .6f\t% .6f\t% .2e\t% .2e\n', ...
            it, x(1), x(2), x(3), err, norm(Fx, inf));
    end
```

```
end

%----------------------------F(x)-------------------------------------%
function F = f_fun(x)
    F = zeros(3,1);
    F(1) = x(1)^3 + x(1)^2*x(2) - x(1)*x(3) + 6;
    F(2) = exp(x(1)) + exp(x(2)) - x(3);
    F(3) = x(2)^2 - 2*x(1)*x(3) - 4;
end

%----------------------------Jacobian---------------------------------%
function J = Jacobian(x)
    J = zeros(3,3);

    % Row 1
    J(1,1) = 3*x(1)^2 + 2*x(1)*x(2) - x(3);
    J(1,2) = x(1)^2;
    J(1,3) = -x(1);

    % Row 2
    J(2,1) = exp(x(1));
    J(2,2) = exp(x(2));
    J(2,3) = -1;

    % Row 3
    J(3,1) = -2*x(3);
    J(3,2) = 2*x(2);
    J(3,3) = -2*x(1);
end
%---------------------------------------------------------------------%
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k       x1           x2           x3          ||dx||_inf      ||F(x)||_inf
-------------------------------------------------------------------------------
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1    -1.636738    -1.514277     0.334707     6.65e-01     1.89e+00
2    -1.464857    -1.668232     0.414166     1.72e-01     1.16e-01
3    -1.456107    -1.664191     0.422475     8.75e-03     6.39e-04
4    -1.456043    -1.664230     0.422493     6.46e-05     1.89e-08
5    -1.456043    -1.664230     0.422493     1.56e-09     1.78e-15

Final solution:
x = [-1.456043, -1.664230,  0.422493]^T
```

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